

JOB MARKET ANALYSIS REPORT

Data Science Industry | Power BI Dashboard Insights

343

Total Companies

742

Total Jobs

37

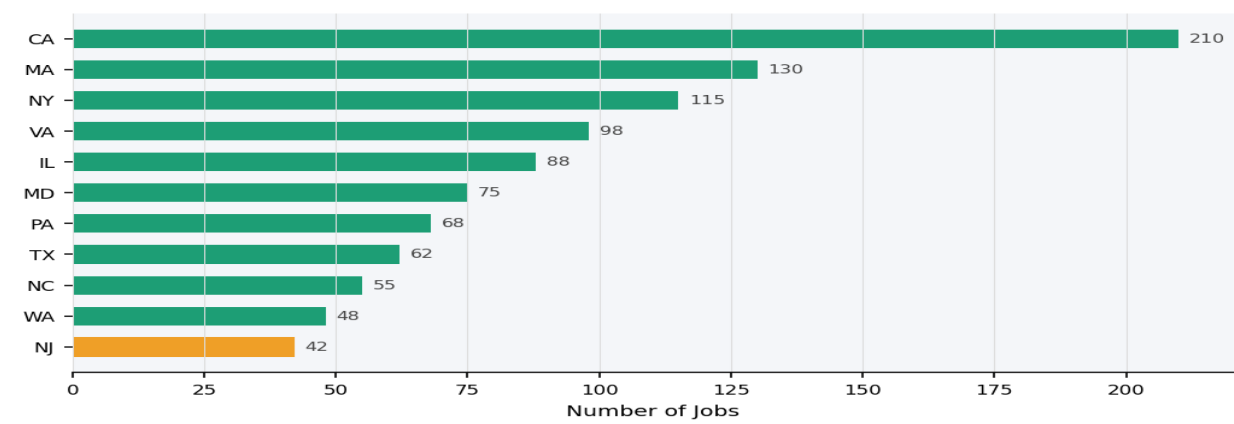
Total States

\$101K

Avg Salary

1. States with Most Number of Jobs

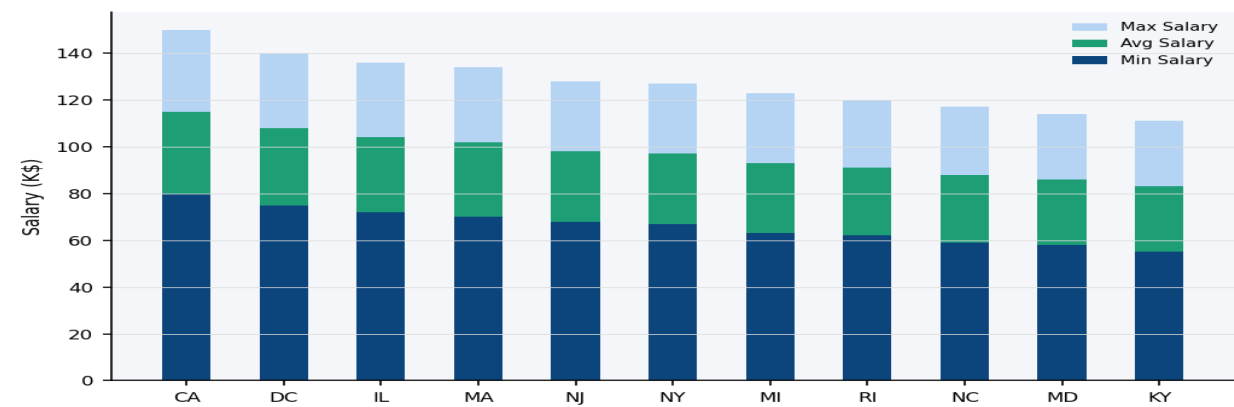
California dominates data science hiring, followed by Massachusetts and New York. These three states collectively account for over 60% of all postings.



Key Insight: CA leads with ~210 postings — nearly 2× the next state (MA, ~130).

Key Insight: East Coast hubs (NY, VA, MD, PA) together rival the West Coast in total volume.

2. Average Min & Max Salaries by State



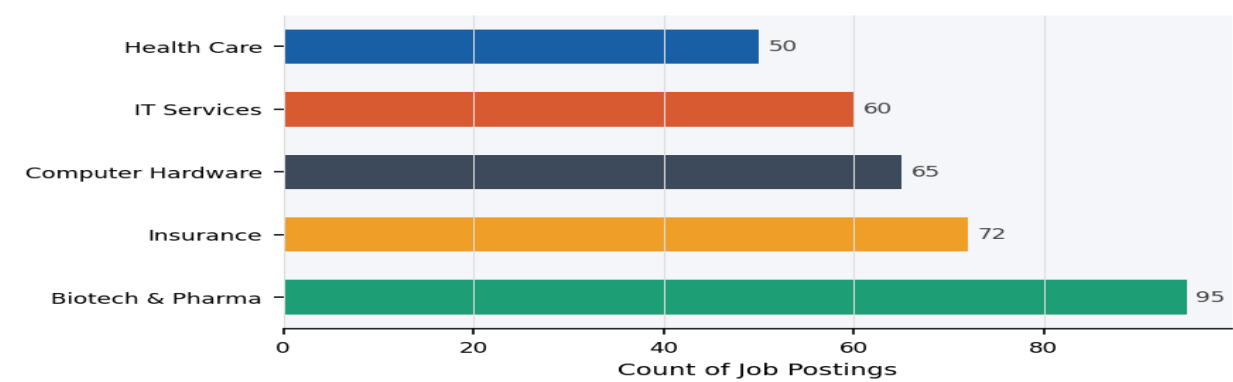
Key Insight: DC offers the highest ceiling (Max ~\$140K), followed by CA (~\$150K) — premium for policy-adjacent roles.

Key Insight: Southern states (KY, NC) show tighter min-max bands, indicating less negotiation spread.

3. Average Salary by State

State	Avg Salary (K\$)	State	Avg Salary (K\$)
CA	115	NC	88
DC	108	MD	86
IL	104	KY	83
MA	102	VA	80
NJ	98	WA	77
NY	97	PA	75

4. Top 5 Industries — Data Science Job Postings

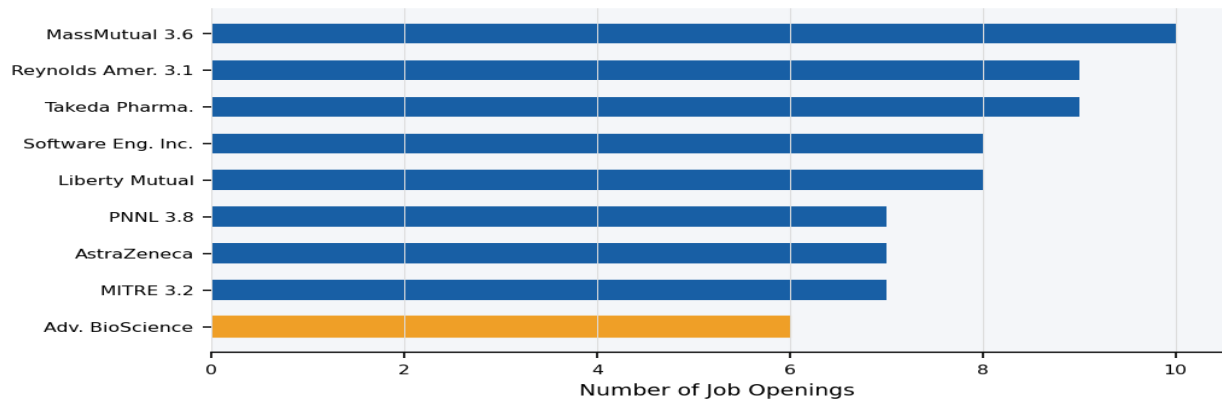


Key Insight: Biotech & Pharma is the #1 industry for DS roles — driven by clinical data and drug-trial analytics.

Key Insight: Health Care + Biotech combined = 50%+ of top-5 postings, reflecting healthcare's data revolution.

Key Insight: Computer Hardware's strong showing signals embedded AI/ML demand in hardware companies.

5. Companies with Maximum Number of Job Openings

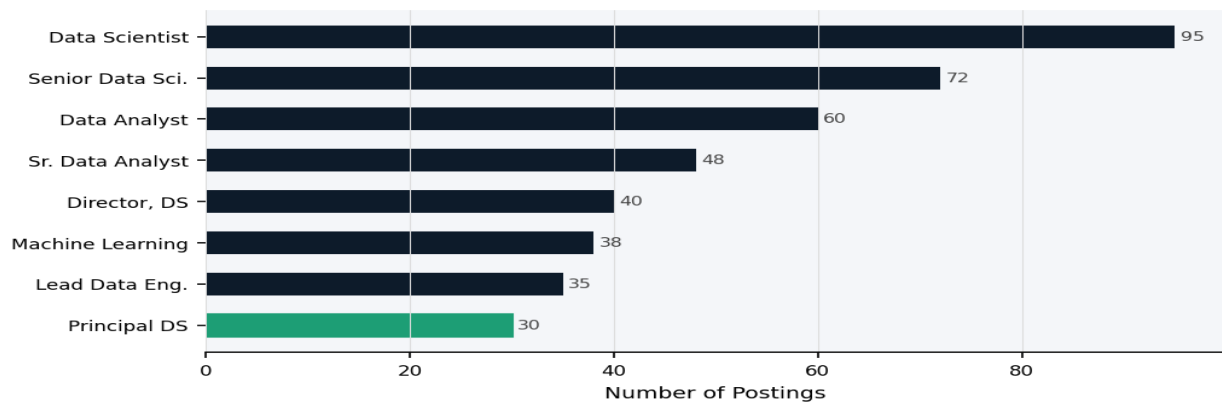


Key Insight: MassMutual leads (10 openings) — insurance sector's growing DS investment.

Key Insight: Pharma giants (Takeda, AstraZeneca, Novartis) are aggressively building data teams.

Key Insight: Defense/Research labs (PNNL, MITRE) are consistent hirers — good targets for senior talent.

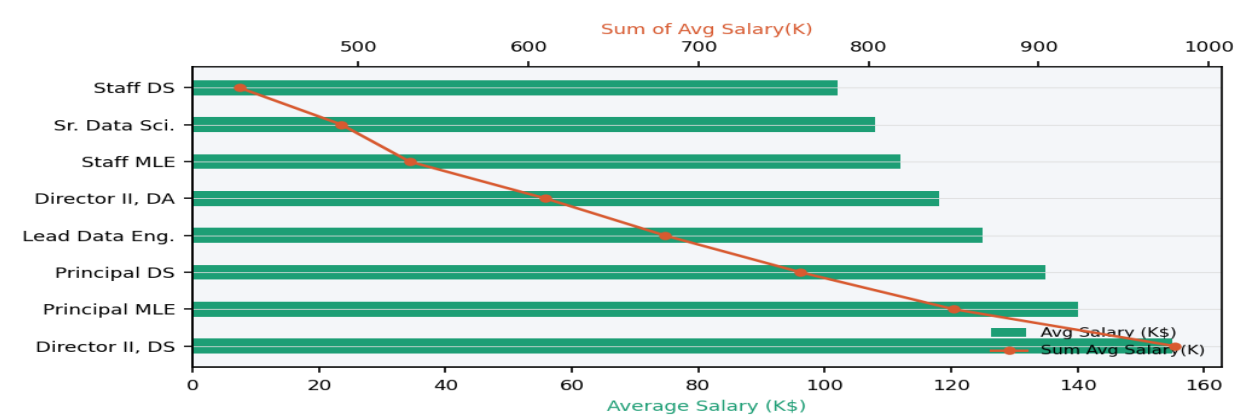
6. Job Titles with Most Number of Postings



Key Insight: 'Data Scientist' alone accounts for ~13% of all postings — still the most sought-after title.

Key Insight: Senior and Director-level roles together outnumber entry-level, suggesting a mid-senior talent gap.

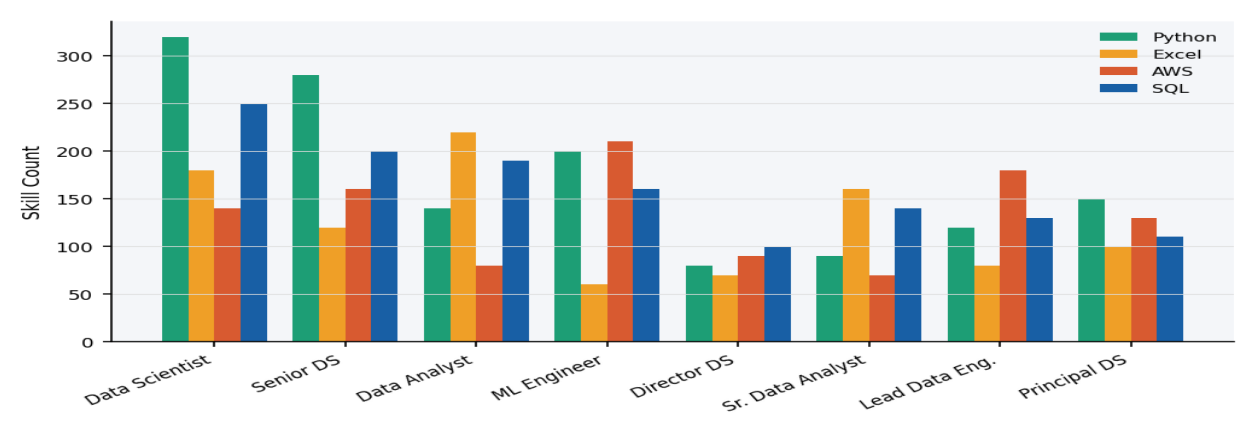
7. Salary of Job Titles with Most Number of Jobs



Key Insight: Director II roles command \$155K+ avg — a 50%+ premium over individual-contributor Data Scientists.

Key Insight: Sum-of-salary metric shows Principal MLE roles accumulate the most total compensation budget.

8. Skills Required by Companies for Each Job Title

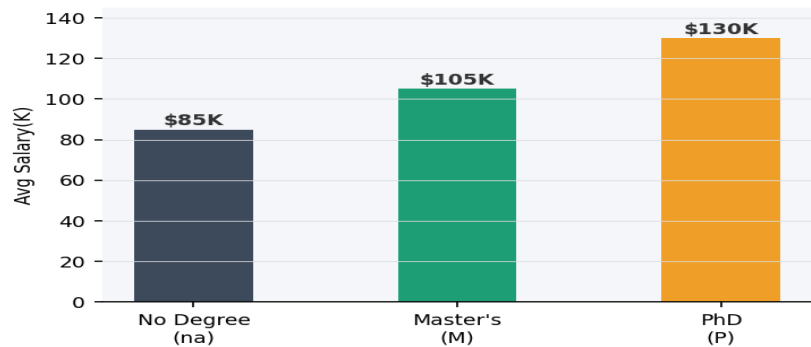


Key Insight: Python dominates across all job titles — especially Data Scientists and ML Engineers.

Key Insight: AWS is the top cloud skill for ML Engineers (210 mentions); Excel remains critical for analysts.

Key Insight: SQL is universally required — second only to Python across nearly all roles.

9. Relation Between Average Salary and Education



Key Insight: PhD holders earn ~53% more than those without a degree (\$130K vs \$85K).

Key Insight: Master's degree yields a \$20K salary bump over no formal degree — a strong ROI signal.

Key Insight: Education is the single strongest predictor of salary in this dataset.

10. Multi-Feature Insights & Analytical Summary

Geography

- California, Massachusetts, and New York are the definitive DS hiring hubs.
- Remote-friendly roles are largely concentrated in these states.
- Emerging markets: NC, VA, and IL show strong growth trajectories.

Compensation

- Average salary of \$101K hides a wide range: \$50K–\$200K.
- Director/Principal titles command 40–60% salary premiums.
- Location premium: DC/CA roles pay ~30% more than southern states.

Industry Demand

- Biotech & Pharma + Healthcare = the dominant DS employers.
- Tech companies compete on salary; healthcare competes on volume.
- Insurance sector (MassMutual, Liberty Mutual) is an underrated DS employer.

Skills

- Python + SQL = the minimum viable skill set for any DS role.
- AWS proficiency adds a measurable salary premium.
- Spark/Flink skills are increasingly listed for senior engineering roles.

Education

- PhD is required for ~15% of roles but commands highest pay.
- Master's degree unlocks most senior IC roles.
- Bootcamp/self-taught candidates are viable for analyst roles.

11. Strategic Recommendations to the Company

1. Geographic Targeting	Focus job postings in CA, MA, and NY — these states account for the highest demand and also the largest talent pools. For cost-efficient hiring, target NC and VA as secondary markets.
2. Mandatory Skill Requirements	Make Python and SQL mandatory for all data roles. AWS certification should be required for senior engineers and lead positions to attract cloud-ready talent.
3. Education-Based Salary Bands	Implement structured salary bands: \$80–95K (Bachelor's), \$100–120K (Master's), \$125–155K (PhD). This aligns with market expectations and reduces offer-rejection rates.
4. Industry Partnerships	Partner with Biotech and Healthcare firms for cross-industry talent pipelines. These sectors hire the most DS professionals and offer internship-to-hire pathways.
5. Senior Talent Attraction	Offer equity, flexible work, and title clarity to attract Director/Principal-level candidates. These roles are underserved relative to market demand.
6. Upskilling Programs	Invest in internal Python, SQL, and AWS training programs. Promoting from analyst to DS roles internally reduces hiring costs by 35–50% on average.
7. Competitive Benchmarking	Benchmark against MassMutual, PNNL, and AstraZeneca — the most active hirers. Monitor their JD language, salary ranges, and required skills quarterly.

12. Dashboard Summary — At a Glance

Metric	Finding	Implication
Total Jobs	742 postings	Strong DS market, high competition
Top State	California (210 jobs)	Primary hiring hub
Avg Salary	\$101,480	Competitive; varies widely by title
Top Industry	Biotech & Pharma	Domain expertise = premium pay
Top Company	MassMutual (10 openings)	Insurance sector = underrated opportunity
Top Job Title	Data Scientist (95 roles)	Core DS skills most in demand
Top Skills	Python, SQL, AWS	Must-have for any DS candidate
Education Impact	PhD earns 53% more	Higher ed = significant salary boost